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Roll No

EC-5005(2)-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Bio Medical Instrumentation

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Assume suitable data, if required.

1. a) Draw and explain the cell and its structure.
b) Write and explain some of the basic components of Bio medical system.
2. a) Explain the operation of Cardio vascular system. Name all the instruments which can be used in this process.
b) What do you mean by transducers? Give a brief classification on transducers.
3. a) Explain the selection criteria of the transducers for any type of bio medical treatment.
b) Explain the operating principle of temperature sensor. Where we can frequently use these types of sensors?
4. a) How you can differentiate normal electrodes with bio instrumental electrodes? Give some points with suitable examples.
b) Give a brief classification of Bio medical electrodes use frequently in the system and explain any two of them.

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5.
 - a) Explain the operating principle of recording methods of data in ECG and EEG of the respective instruments.
 - b) Explain the principle of the instrument use to measure blood pressure. What are the measures we have to decide before operate the machine?
6.
 - a) Discuss some of the instruments use for checking safety parameters of bio medical equipment's.
 - b) Blood gas analyzers are used with some safety measures. What are they? Briefly explain each one of them.
7.
 - a) Discuss the parameters to be recorded in Radio graphic machines.
 - b) Give a brief introduction of Biometric System.
8. Write short notes (any four):
 - a) Defibrillators
 - b) Lithotripsy
 - c) Computer Tomography
 - d) Photoplenthysmography
 - e) Piezoelectric Effect
 - f) Shock Hazards Instruments

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